Run all the tools at the same time

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This page explains how to run the three real tools together in one command, what each one tests, and what the saved results mean.

> **Permission first.** Nmap and sqlmap run against public sites, and each site is used only for the tool its operator allows. Metasploit runs only on your own computer (`127.0.0.1`). No public site permits Metasploit testing.

1. What the command runs

The command starts **three separate engagements at the same time**. Each engagement runs its own tool.

Engagement	Tool	Target	Kind of target	Permission in this project
1	Nmap	`http://scanme.nmap.org/`, ports 22, 80, 443	Public	Operator permits Nmap only, about 12 scans per day, no exploits or denial of service
2	sqlmap	`https://sekumart.sekuripy.hr/product.php?id=1`, parameter `id` only	Public practice site	Operator's SQL injection playground; sqlmap only
3	Metasploit	Temporary listener on `127.0.0.1` (your computer)	Local lab	Only the reviewed port scanner `auxiliary/scanner/portscan/tcp`

"At the same time" means the three engagements run concurrently. Each engagement still runs its own tool one step at a time. Results are not shared between engagements.

The command never accepts a target from you. The public targets are fixed in the code, and the command refuses to run without the confirmation flag.

2. Before you start

1. The project's Python environment exists at `backend/.venv` (see the beginner guide).
2. **Nmap** is installed (`nmap --version` works).
3. **sqlmap** is installed. Note its path (on Windows, the `sqlmap.py` path from the clone).
4. **Metasploit (optional but required for engagement 3):** the real `msfrpcd` daemon is running on `127.0.0.1` port `55553`. See **metasploit how to use.pdf**, sections 2 and 3. If it is not running, engagement 3 reports **FAILED**. It does not report a pass.
5. The daemon password is in your private `backend/.env` or set in the terminal. Do not print it or paste it into chat.

3. Commands

Run these from the **project root**, in the VS Code terminal.

Step 1: set the terminal values (this window only)

Mac (zsh):

```
export MSF_RPC_PASSWORD="YOUR_STRONG_PASSWORD"
export MSF_RPC_PORT=55553
export SQLMAP_PATH="$(command -v sqlmap)"
```

Windows (PowerShell):

```
$env:MSF_RPC_PASSWORD = "YOUR_STRONG_PASSWORD"
$env:MSF_RPC_PORT = "55553"
$env:SQLMAP_PATH = "$HOME\sqlmap\sqlmap.py"
```

Use the same password as the `-P` value you gave `msfrpcd`. On Mac, `sqlmap` installed with Homebrew is found automatically; the `SQLMAP_PATH` line is optional there. On Windows, change the path if your clone is elsewhere.

Step 2: run all three at the same time

Mac (zsh):

```
./backend/.venv/bin/python scripts/live_tests/run_all_tools.py --confirm-public-tests
```

Windows (PowerShell):

```
.\backend\.venv\Scripts\python.exe scripts\live_tests\run_all_tools.py --confirm-public-tests
```

The run takes about 30 seconds to 2 minutes. Each line starts with the engagement name and shows `CAPTURED` (passed) or `FAILED`.

Expected:

```
Starting three engagements at the same time: Nmap, sqlmap, Metasploit...
nmap: CAPTURED — The real tool completed and the framework returned a recognizable result.
sqlmap: CAPTURED — The real tool completed and the framework returned a recognizable result.
metasploit: CAPTURED — The real Metasploit daemon scanned the local listener and reported the open
port.
```

Step 3: read the results

The run saves:

- `docs/all_tools_result/run-all-tools.json`: each engagement's status, time, target, version, and output excerpt
- `docs/real_results.json` and `docs/real result.pdf`: the latest results, which are rebuilt automatically

Copies from the 2 November run are in **Real results / all tools result**.

Step 4: after the run

Stop `msfrpcd` with `Ctrl+C` in its own window. Clear the variables by closing the terminal.

4. Limits on public tests

- Public scans count toward the operator limits. Nmap is limited to about 12 scans per day, and the command enforces 10 per rolling 24 hours. sqlmap is limited to 6 per 24 hours by the command.
- Do not run this command repeatedly. Run it once when you need a fresh capture.
- The command refuses to run without `--confirm-public-tests`.

5. Results from the run on 2 November 2026

Engagement	Status	Started (UTC)	Finished (UTC)	Result
Nmap · `scanme.nmap.org`	**CAPTURED**	14:14:36	14:14:54	Nmap 7.95. Ports 22 (OpenSSH 6.6.1p1), 80 and 443 are open.
sqlmap · SekuMart `id`	**CAPTURED**	14:14:36	14:14:43	sqlmap 1.10.11.2#dev. Injection found in GET `id`, error-based, back-end MySQL. Detection only; no data was dumped.
Metasploit · `127.0.0.1` lab	**CAPTURED**	14:14:36	14:14:36	Metasploit 6.5.9 (real daemon). Found the open port 39873 that the lab listener was using.

The three engagements started at the same second, so they ran at the same time.

Note on the first attempt: an earlier run of the same command reported Metasploit as failed. The cause was a wrong field name in the new script (`tool` instead of `msf_module`). The Metasploit result was correct, so the script was fixed and run again. The table above shows the corrected run.

6. What this shows and what it does not

Shows: - Nmap, sqlmap and the real Metasploit daemon all run, and their outputs are read back into Cerberus. - Three engagements can run at the same time without interfering with each other. - Out-of-scope requests are refused before any tool runs (tested separately).

Does not show: - That any public website is vulnerable to anything beyond the specific, documented results. - Metasploit's behavior on a public site. No public site permits it. - That Mac or Windows behaves the same way. These steps were run on Linux only. - A finished product. The API, AI agent and interface do not start these tools yet.

7. Troubleshooting

If you see	Try this
`nmap executable is missing`	Install Nmap, then reopen the terminal.
`sqlmap executable is missing`	Set `SQLMAP_PATH` for this window (Step 1).
`metasploit: FAILED` with a connection error	`msfrpcd` is not running, or the port is not `55553`. Start it and check Step 1.
`metasploit: FAILED` with an authentication error	The password in Step 1 does not match the `-P` value of `msfrpcd`.
Metasploit times out	Check that `msfrpcd` is running and that the port in Step 1 matches it. Retry once. On the test machine, creating the console took under one second.
`public tests require --confirm-public-tests`	Add the flag to the command.
Rate limit reached	Wait for the 24-hour window to pass. The command refuses before scanning.
PDF could not be rebuilt	The results are saved. Install `backend/requirements-dev.txt` and run `scripts/render_real_results.py`. Do not rescan.

8. Files

File	What it is
`scripts/live_tests/run_all_tools.py`	The runner
`backend/tests/unit/test_run_all_tools.py`	Tests that the runner refuses to start without confirmation and uses the approved targets
`docs/all_tools_result/run-all-tools.json`	Saved output of the run